4 Fabrication Process Techniques

Oxidation

Oxidation is the process by which a layer of silicon dioxide is grown on the surface of a silicon wafer. Its uses are:

1. **Surface Passivation**
   SiO_2 layer protects SC devices from contamination. SiO_2 performs a role in two ways.
   a. **Physical protection:** this entails the physical protection of the surface and underlying devices. SiO_2, like glass, is a hard and durable material. The hardness protects the wafer surface from the surface and abuse (shocks) during the fabrication process.
   b. **Chemical protection**: some electrically active contaminants (mobile ionic contaminants) that end up on the wafer surface are drawn to SiO_2 during thermal oxidation and become less harmful to the devices.
2. **Doping Barrier**

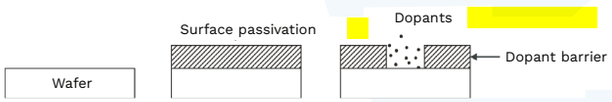


Fig. 4.1

3. **Surface Dielectric**

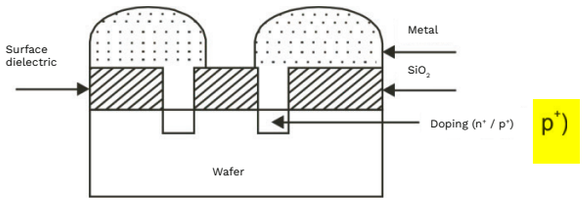


Fig. 4.2

   a. Thick oxide prevents charges induced by electrical currents flowing in the metal layers into the wafer surface.
   b. An oxide that covers most of the wafer surface is called field oxide.
4. **Device Dielectric**
Thin layers of SiO_2 are grown in the gate regions.

SiO_2 thickness (A°)	Application
60–100	Tunnelling gates
150–500	Gate oxides and capacitor dielectrics
200–500	LOCOS pad oxide
2000–5000	Masking oxides and surface passivation
3000–10000	Field oxides

Table 4.1

Thermal Oxidation Mechanism

- Using this mechanism, three methods are used—atmospheric growth, rapid thermal oxidation, and high-pressure oxidation.
- In all methods, the basic mechanism of growth is the same.
  $Si\ (Solid) + O_2\ (gas) \rightarrow SiO_2\ (Solid)$
- The above reaction occurs even at room temperature. Elevated temperature is required to achieve quality in oxides.
- Oxidation temperatures are between 900°C and 1200°C.
- Actual growth happens in two stages, i.e., linear and parabolic.
- In linear, oxide grows in equal amounts for each unit of time. After approximately 500A° grown, a limit is imposed on the growth rate.
- The growth rate limit is a result of the separation of Si surface from the oxygen vapours by the oxide layer grown in the initial stage.

- For oxide growth to continue, O_2 migrates (diffuses) through the SiO_2 layer to the Si wafer surface. Thus, the layer of SiO_2 grows into the Si surface.
- $R = \frac{x^2}{t}$, R is the oxidation rate, X^2 denotes oxide thickness, and t denotes time.
- Parabolic growth is also called transport limited reaction.

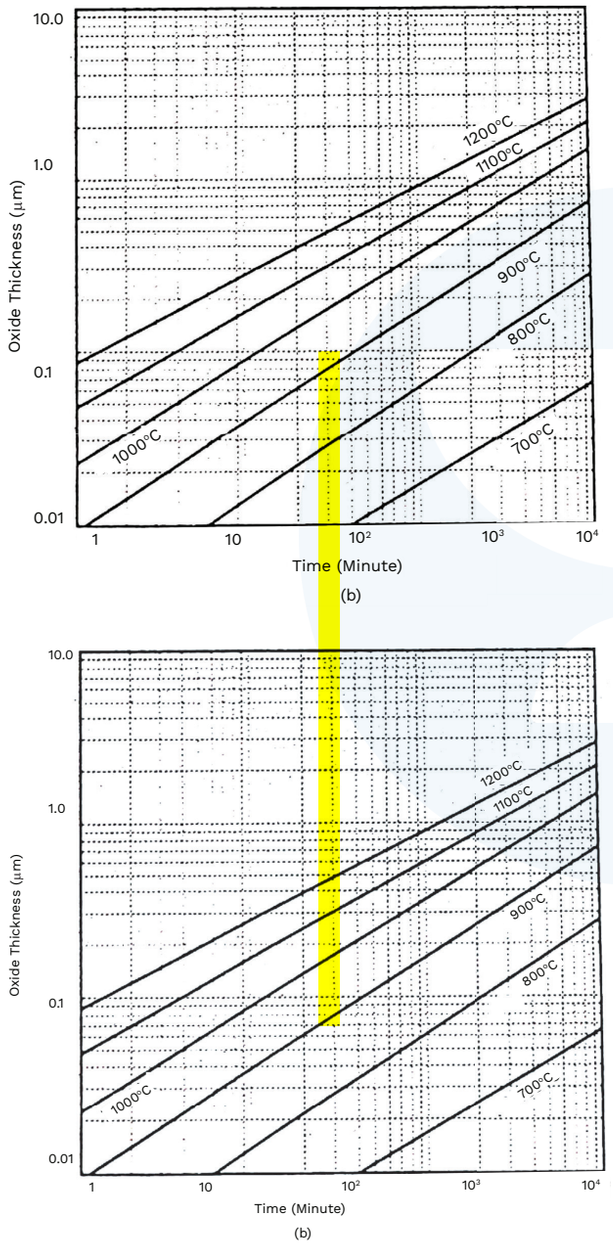


Fig. 4.3 Silicon Dioxide versus Time for Different Temperatures and Orientation in (a) Dry Oxidation and (b) Steam

Dry and Wet Oxidation

- Dry: slow but good quality oxide is grown by this method.
  Wet: the rate of oxidation is faster but the oxide formed is lower in quality as compared to dry oxidation.
- H_2O (Water vapour) ($H\text{-}OH^-$), OH^- (Hydroxyl ion) diffuses faster through oxide layers to reach Si surface. So, wet oxidation is faster than dry oxidation.
- After heating the oxide in an inert atmosphere, such as nitrogen, the two oxides become identical in structure and properties.
- Steam oxidation, wet oxidation, and pyrogen steam are all same.
- $Si\ (Solid) + H_2O\ (gas) \rightarrow SiO_2 + 2H_2\ (gas)$
  It makes the oxides less dense than the oxide grown in dry oxidation.

Thermal Oxidation Methods

In silicon technology, the energy required for oxidation is usually supplied by heating the wafers called thermal oxidation. Oxide layers are grown either at atmospheric pressure or at high pressure. There are two atmospheric techniques, tube furnaces (horizontal and vertical tube furnaces) and rapid thermal systems.

Rapid Thermal Processing (RTP)

Ion implantation has replaced thermal diffusion due to its inherent doping control. However, ion implantation requires a follow-on heating operation called annealing to cure crystal damage induced by the implant process. Annealing causes the dopant atoms to spread out in the wafer, which leads to an undesirable result. So, to overcome this problem, RTP technology was developed. In RTP technology, short heating and cooling times keep the implanted atoms to stay in their original location.

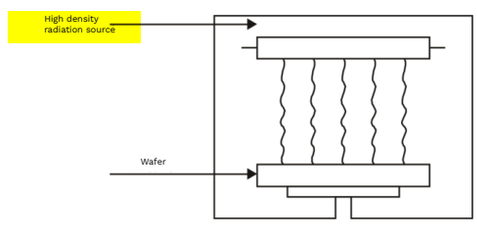


Fig. 4.4 RTP

High Pressure Oxidation

The problem of the growth of dislocations in the bulk of the wafer and the growth of hydrogen induced dislocations on the surface of the wafer led to the introduction of high pressure oxidation systems.

In this system, the oxidant is pumped into the tube at the pressures of 10 to 25 atm (10 to 25 times the pressure of the atmosphere). At these pressures, oxidation proceeds at a faster rate than in atmospheric systems.

An increase in pressure by 1 atm allows a 30°C drop in temperature. In a high pressure system, that high pressure relates to a drop of 300° to 750°C in temperature and this reduction is sufficient to minimise dislocation growth in and on the wafer.

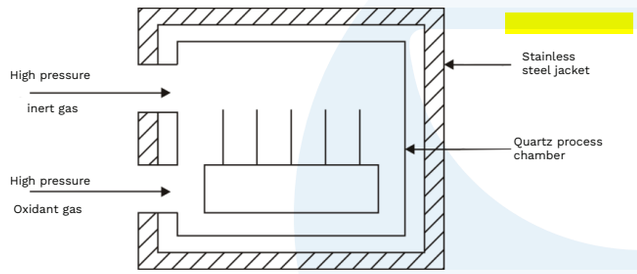


Fig. 4.5 High Pressure Oxidation

Anodic Oxidation

In addition to thermal oxidation, there is one chemical oxidation process to grow thin oxides on silicon. This method is called anodic oxidation.

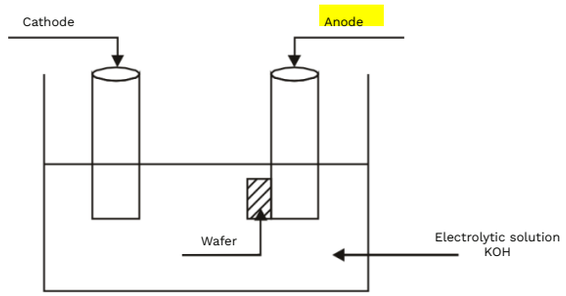


Fig. 4.6 Anodic Oxidation

- The quality of oxide formed in this process is lesser than thermally grown oxide.
- It is used for the formation of thin gate oxides.
- In anodic oxidation, silicon from the silicon surface travels to the top of the oxide rather than oxygen travelling to the wafer surface.
- When anodic oxide is etched from a silicon surface, it leaves dopants near the surface undisturbed, unlike thermally grown oxide.

Lithography

- Lithography is the patterning process that removes specific portions on the top layer(s) on the wafer surface.
- It is a two-step process.
  - Create, in and on the wafer surface, a pattern whose dimensions are as close to the design requirements. This goal is referred to as resolution of the images on the wafer. The pattern dimensions are referred to as feature or image sizes of the circuit.
  - There is alignment or registration (correct placement) of the circuit pattern on the wafer.
- The final pattern is created from several photo masks applied to wafer sequentially.
- Registration is similar to the correct stacking of different building floors in relation to each other.
- Misaligned mask layer leads to the failure of the entire circuit.

Process

- Lithography is similar to stencilling.
- The required pattern is first formed in reticles or photomasks, and then, transferred to the wafer surface through photomasking steps.
- The transfer takes place in two steps.
  - Pattern on the mask is transferred onto the layer of photoresist (PR). Photoresist is a light sensitive material that changes its structure and properties when exposed to light.

- The pattern is transferred from the PR layer to the wafer surface. The transfer occurs when the etchants remove the portion of the wafer's top layer not covered with PR (PR is etch-resistant).

Positive (+ve) and Negative (–ve)

+ve PR: photo solubilisation takes place (after exposure, insoluble becomes soluble).
–ve PR: polymerisation takes place [after exposure, soluble becomes insoluble (hard)].

Masks Light and Dark Field



Fig. 4.7 Mask Reticles



Fig. 4.8 Photo Masking Hole and Island (After Exposure)

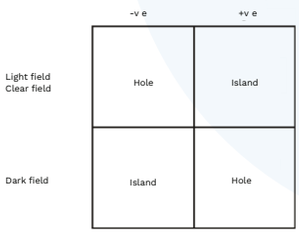


Fig. 4.9 Mask and PR Result

First Transfer (Mask to PR)

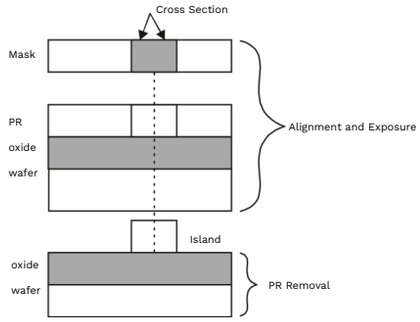


Fig. 4.10

Second Transfer (PR to Wafer)

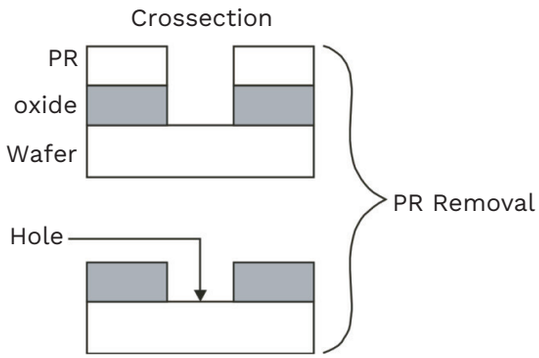


Fig. 4.11 Image Transfer Using –Ve Pr and Light Field Mask

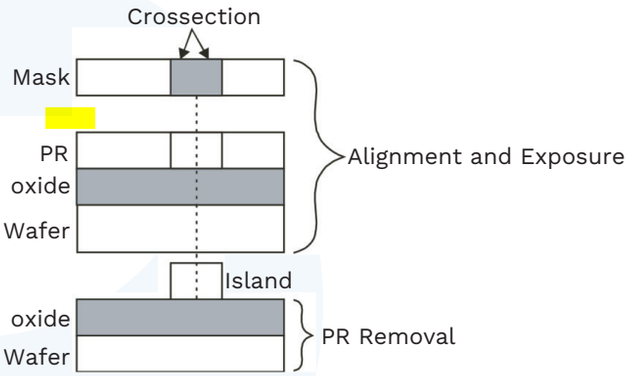


Fig. 4.12 Image Transfer from a Light Field Mask to a +ve PR

- –ve PR generally has higher adhesion capability than +ve resists.
- –ve PR requires 5–15 seconds of exposure time, +ve PR takes three to four times longer.
- Wafer goes through a chemical priming step to ensure good adhesion of the resist.
- Hexamethyldisilazane (HMDS) is in universal use as a primer.

PR Spinning

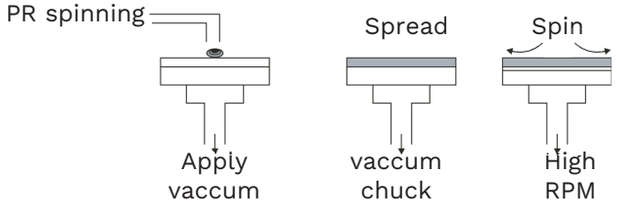


Fig. 4.13 PR Spinning

Alignment and Exposure Systems

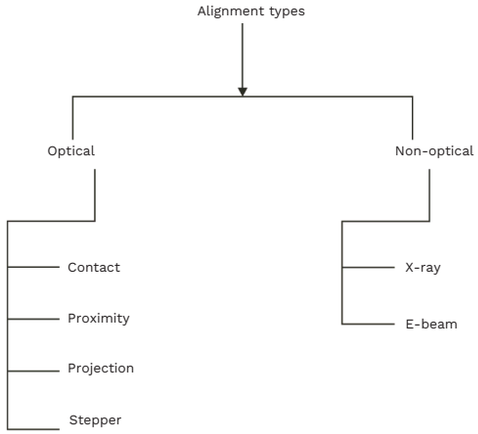


Fig. 4.14

- Optical aligners use UV light source.
- Non-optical aligners use electromagnetic spectrum.

Misalignment

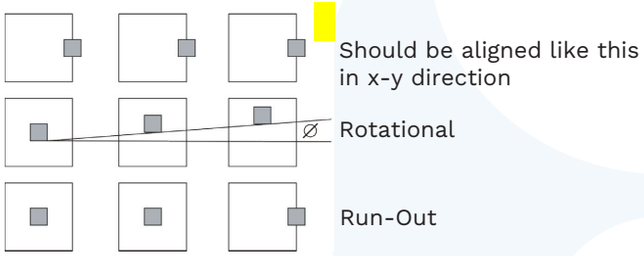


Fig. 4.15 Misalignment

Contact aligners: the mask is in intimate contact with the wafer.

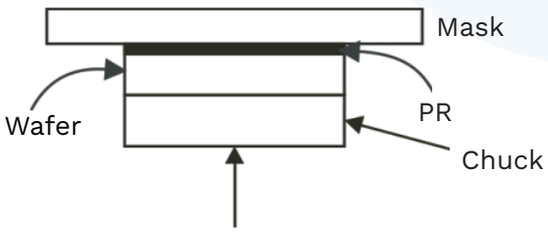


Fig. 4.16 Contact Aligner

Proximity aligners: it was the evolution of contact aligners. The wafer is kept near or in soft contact with the mask. It reduces the number of defects.

Scanning projection aligners: there is projection of mask image onto the wafer surface.

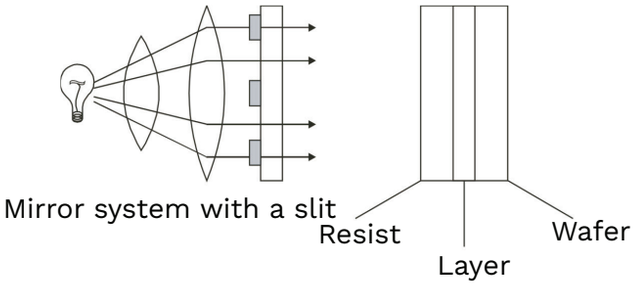


Fig. 4.17 Scanning Projection Aligner

Steppers: it involves stepping up and repeating the aligner. There is resolution improvement because a smaller area is exposed each time and it is less vulnerable to dust and dirt.

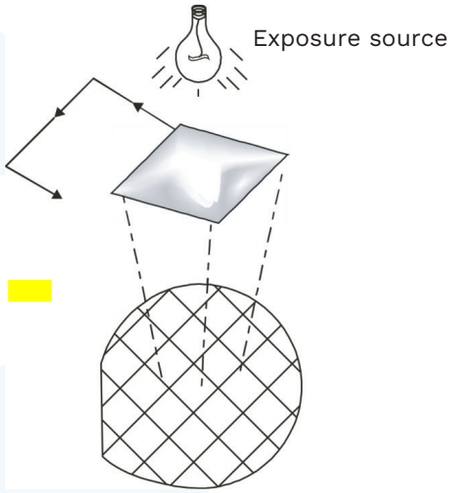


Fig. 4.18 Stepper

X-ray Aligners

- High output through short exposure times.
- There is a low level of defects and dust.
- It has a higher resolution exposure source.

E-beam Aligners

- It is used in the production of high quality masks and reticles.
- It is costly.
- Its speed is slow.

Resolution (Microns)	Contact	Projection	Stepper	X-ray	E-beam
	3.0 – 5.0	1.5 – 2.0	1.5 – 2.0	0.3 – 0.5	0.75 – 1.0
Throughput (Wafers/Hr.)	80	50 – 60	30 – 40	20	2

Table 4.2

Development: after the wafer completes the alignment and exposure step, the device or circuit pattern is coded in the photoresist as regions of exposed and unexposed resist. The pattern is 'developed' in the resist by the chemical dissolution of the unpolymerised resist regions.
Development is the technique designed to leave in the resist layer an exact copy of the pattern that was on the mask or reticle.
Negative and positive resists have different developing characteristics and require different chemicals and processes.

	Positive	Negative
Developer	NaOH Tetramethyl Ammonium Hydroxide (TMAH)	Xylene Stoddard solvent
Rinse	H_2O	n-Butylacetate

Table 4.3 Resist Developer and Rinse Chemicals

The development step is followed by a rinse to:
1. rapidly dilute the developer chemical to stop developing action.
2. remove partially polymerised pieces of resist from the open regions in the resist film.

Etching

After completing the development step, the mask (reticle) pattern is defined in the photoresist layer and is ready for etching. During the etch step, the image is permanently transferred to the surface layer on the wafer.
The ideal etch leaves vertical sidewalls in the layer and this ideal result is called anistropic etch. But if the result is a hole wider at the top than at the bottom with a sloped sidewall, then this unwanted result is called isotropic etch.

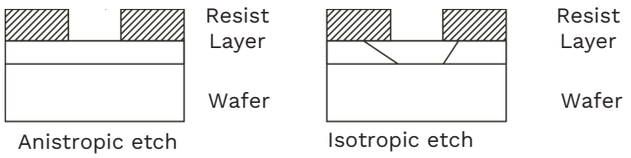


Fig. 4.19 Anisotropic and Isotropic Etch

Two Goals of Etching

1. It helps in controlling undercutting (undercutting action of etching chemical causes isotropic etch). Undercutting occurs due to excessive etch time, high etching temperature, or if etch mixture is too strong.
2. It helps in preservation of the surface underlying the etched layer. This preservation of the surface is called selectivity.

Etching Methods

There are two basic techniques of etching—wet and dry etching.

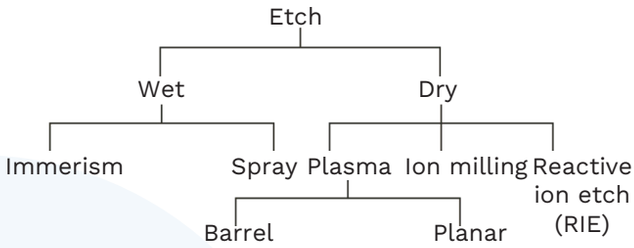


Fig. 4.20 Etch Methods

Wet Etching

For wet etching, wafers are loaded into an etch-resistant boat and immersed in a tank of the etchant. After a predetermined time in the etch tank, they are processed through the rinsing and drying steps.

Common Etchants	
Si	Solution of nitric and hydrofluoric acid in wafer
SiO_2	Hydrofluoric acid (HF acid) Ammonium fluoride (NH_4F)
Aluminium	Phosphoric acid + Nitric acid + Acetic acid + Water
Silicon Nitride (Si_3N_4)	Phosphoric acid

Table 4.4 Common Wet Etchants Used for Etching Silicon, Silicon Dioxide, Aluminium, and Silicon Nitride

Wet Spray Etching

It is advantageous over immersion methods in the following ways.
1. Added definition is gained from the mechanical pressure of the spray.
2. It minimises contamination from etchants.

3. It can be more controllable as etchants can be instantly removed from the surface by switching the system to a wafer rinse.

It has a few disadvantages also like system cost and safety considerations associated with caustic etchants in a pressurised system.

Dry Etching

The limits of wet etching are as follows.

1. It is limited to pattern size of 3 μm.
2. It requires rinse and dry steps.
3. Wet chemicals are hazardous and toxic.
4. It has a potential for contamination.
5. Undercutting

The above considerations led to the use of dry etch processes.

In dry etching, gases are the etching medium. Wafers enter and exit the system in a dry state.

Barrel Plasma Etching

It is a chemical process that uses plasma energy to drive the reaction.

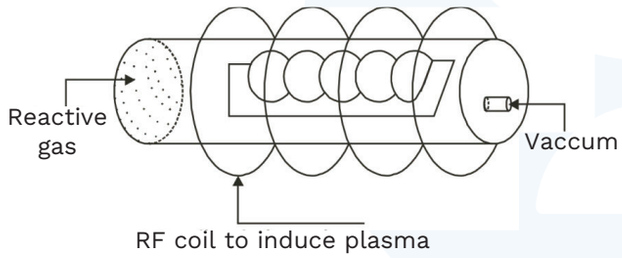


Fig. 4.21 Barrel Plasma Etch

Earlier, plasma systems were designed with circular chambers. These were called barrel etchers. In this system, uniform etching is difficult because of the non-directional of etching ions, which led to the use of planar systems.

Planar Plasma Etching

In this system, etching ions are more directional than those in the barrel system. The result is a more anisotropic etch. Uniformity increases with the rotation of the wafer pallet (grounded pallet) in the system.

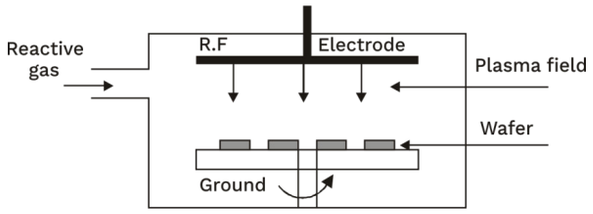


Fig. 4.22 Planar Plasma Etch

Ion Beam Etching

It is a physical process.

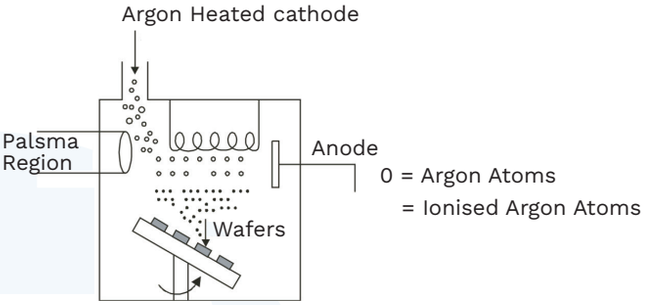


Fig. 4.23 Ion-beam Milling

Wafers are placed on a vacuum chamber holder and argon steam is introduced into the chamber. The electrons from a set of cathode–anode electrodes ionise the argon atoms with a +ve charge. At the wafer surface (wafers are held on a negatively charged holder), ionised argon atoms crash and blast small amounts from the wafer surface. This action is called momentum transfer, which is a physical process. This method is also called sputter etching or ion milling.

Etching is highly directional, resulting in a good definition of small openings. Two considerations of this method are its poor selectivity and radiation damage due to the ionisation mechanism.

Reactive Ion Etching

It combines plasma and ion-beam etching principles. The systems are similar in construction to plasma and ion-beam systems but have ion-milling capability. The combination brings the benefits of both plasma and ion-milling etching.

A major advantage of RIE systems is in the etching of silicon dioxide over silicon layers. It results in selectivity ratio of 35:1, whereas it is only 10:1 in plasma etching.

Diffusion and Ion Implantation

1. Two techniques used to create conductive layers and p–n junctions in the wafer surfaces are diffusion and ion implantation.

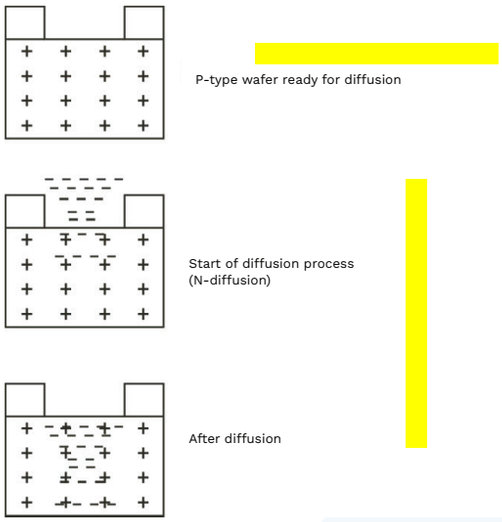


Fig. 4.24

Layer	N	P	(N – P)	Layer
1	12	5	7	N
2	10	5	5	N
3	8	5	3	N
4	5	5	0	jn
5	3	5	–2	P
6	0	5	–5	P

Table 4.5 Dopant Amounts and Level Conductivity Type

2. Junction is the Interface between p and n-type regions in a semiconductor. It is the location (jn) where a number of n-type dopants is equal to p-type dopant atoms (5-5 in the above table at layer number 4).

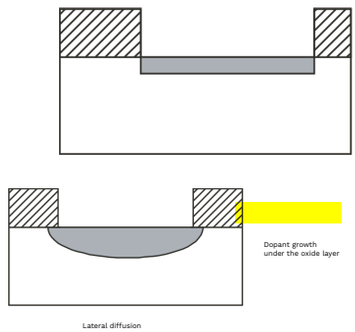


Fig. 4.25 Crossectional Representation of Diffused Junctions

Concept of Diffusion

- The movement of one material through another is a natural chemical process called diffusion.
- Two conditions are necessary for diffusion to take place.
  - One of the materials should be at a higher concentration than the other.
  - There should be sufficient energy in the system for high concentration to move into the other.
- The action of spray in a room, use of fragrance sticks, mosquito coil, etc., are examples of diffusion.

Solid-State SC Diffusion

1. A complete diffusion process requires two steps called deposition and drive-in oxidation.
2. The goals of a diffusion process are threefold.
   a. Creation of a specific concentration of dopant atoms in the wafer surface.
   b. p–n junction at a specific distance below the wafer surface.
   c. Specific distribution of dopant atoms in the wafer surface.
3. The movement of dopants takes place by two mechanisms.
   a. Dopant atom moves by filling empty crystal positions, i.e., the vacancies.
   b. Dopant atom moves through the spaces between the crystal sites, i.e., i-e intersite.

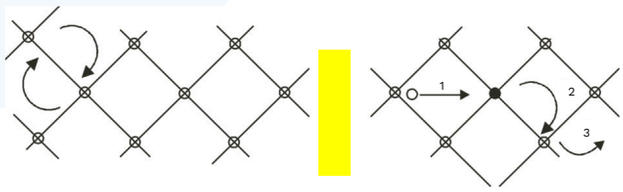


Fig. 4.21 : (a) Vacancy model (b) Interstitial model

Fig. 4.26

4. Deposition is controlled by several factors.
   a. One among them is the diffusivity of the particular dopant.
      i. It measures the rate of movement of the dopant through the particular wafer material.
      ii. Diffusivity increases with temperature.
   b. Another factor is maximum solid solubility of the dopant in the wafer material.

i. It is the maximum concentration of a particular dopant that can be put into the wafer (like the maximum liquid solubility of sugar in coffee).
ii. Its limit also increases with temperature.

Dopant Sources (Solid, Liquid, and Gas)

- **Liquid**
  - Advantage: low to moderate cost
  - Disadvantage: uniformity problems (especially for large diameter wafers)
- **Gas**
  - Advantage: precise control and favourable for large diameter wafers
  - Disadvantage: contamination due to silica dust
- **Solid**

  Cost, uniformity, and breakages are different for different solid sources.

Drive-in Oxidation

The second major part of the diffusion process is the drive-in oxidation step. It is also called reoxidation or re-ox. It has a twofold purpose.

1. Redistribution of the dopant is deeper into the wafer.
   a. In drive-in, heat drives the dopant atoms deeper and wider into the wafer.
   b. The junction depth increases.
   c. Drive-in oxidation steps take place at high temperatures than the deposition step.
2. There is oxidation of the exposed si-surfaces.

Introduction to Ion Implantation

1. With the advent of VLSI and MOS based circuits and their smaller feature sizes, diffusion become a processing limitation.
2. There are four problems of diffusion.
   a. Side diffusion (lateral diffusion)
   b. Low doping control
   c. Surface doping
   d. Dislocation generation
3. Lateral diffusion continues whenever the wafer is processed at a high temperature.
4. There is difficulty in creating doped regions of low concentration.
5. Majority of the dopant atoms are located near the wafer surface.
6. Dislocations occur at high temperature.

Ion implantation overcomes these limitations of diffusion in the following ways.

- There is no side diffusion.
- The process takes place close to room temperature.
- Dopant atoms are placed below the wafer surface.
- A wide range of doping concentration is possible.
- Diffusion is still used in low integration level circuits.

Concept of Ion Implantation

1. It is a physical process and not a chemical process.
2. Dopant atom sources are ionised, selected, accelerated, and formed into a beam. The dopant atoms enter the wafer and come to rest below the surface.

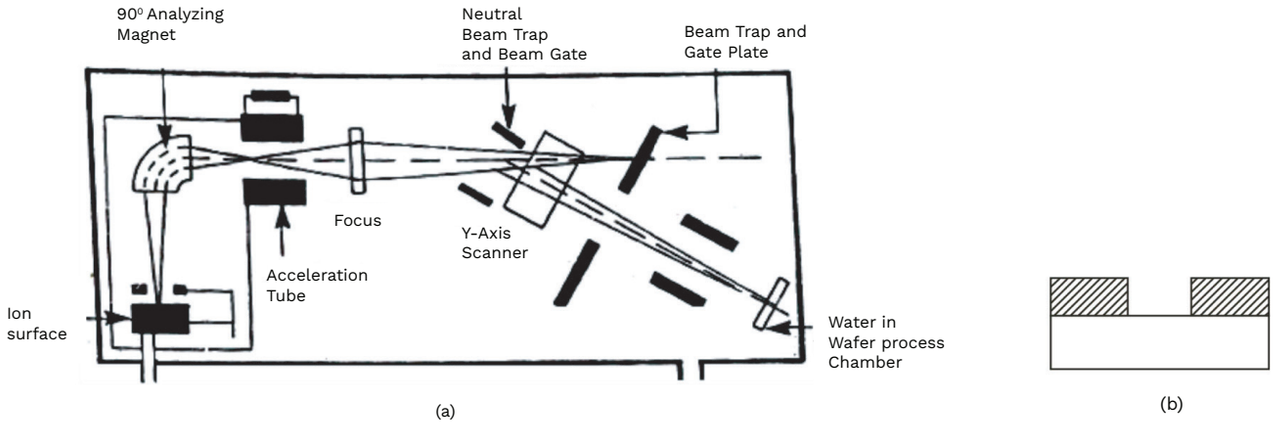


Fig. 4.27 Ion Implantation (a) Block Diagram of Ion implanter, and (b) Distribution of Implanted Atoms in a Wafer

Implant Sources

- It includes only gas and solid.
- Gases are favoured because of their ease and higher control.
- Gases are mostly fluorine based, such as pF_5, AsF_5, $B^{11}F_3$, SbF_3, and PF_3.
- Certain solid sources are used for special applications. One of them is phosphorous pentoxide (P_2O_5).
- Powder is heated to create vapours carried into the ion source chamber.

Ionisation Chamber

1. Ions are atoms or molecules with a –ve or +ve charge. The ions implanted are ionised atoms of the dopants.
2. The chamber is maintained at low pressure (vacuum of about 10^{-3} torr).
3. Inside the chamber is a filament that is heated to the point where electrons are created from the filament surface. Electrons are attracted to an oppositely charged anode in the chamber. During the travel from the filament to the anode, the electrons collide with the dopant source molecules and create a host of positively charged ions from the element in the molecule.
4. Another ionisation technique uses a cold cathode technique to generate electrons.
5. A high voltage EF is created between the cathode and anode, which creates the electrons in a self-sustaining process.

Ion Selection

1. Top of the list is a lone boron ion.
2. Other species cannot be implanted.
3. The boron ion is selected from the group by analysing the selection of separation.
4. The analyser creates a magnetic field.
5. In the field, each of the positively charged species is bent into an arc with a specific radius.
6. At the exit end of the analyser is a slit that allows only boron ion to exit. Thus, only boron ions leave the analysing subsystem.

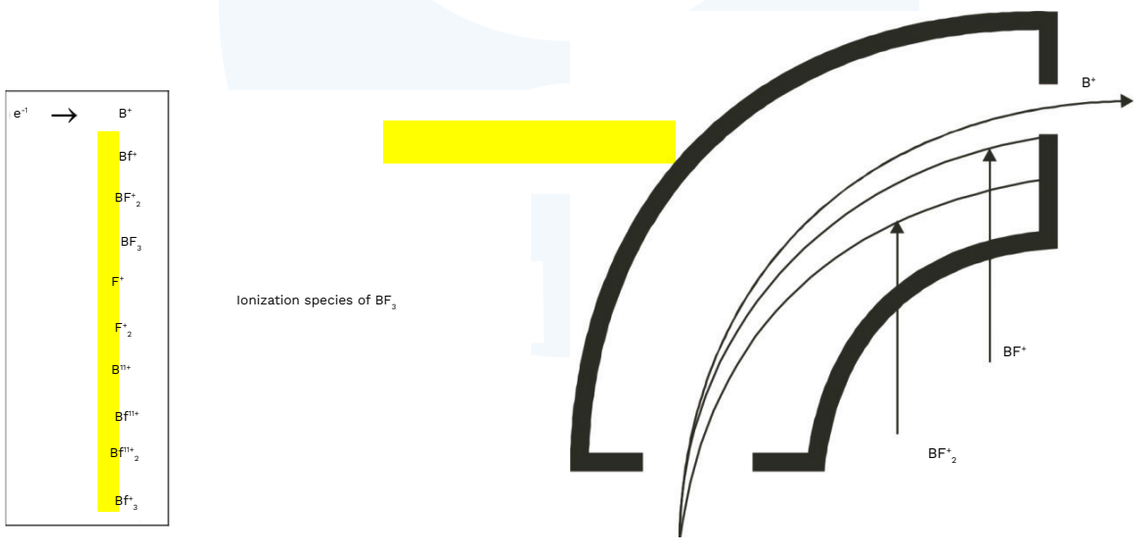


Fig. 4.28 Analysing Magnet

Acceleration Tube

- Accelerate the ion to a high enough velocity so that it will have the momentum to penetrate the wafer surface.
- Higher the voltage, the higher is the momentum.
- The amount of atoms implanted is called dose.

Beam Focus

The beam separates due to the repulsion of like charges, so electrostatic or magnetic lenses are used to focus the ion into 1 cm diameter beam.

Neutral Beam Drop

- Despite the vacuum removal of the majority of the air, there are still some residual gas molecules in the vicinity of the ion beam.
- Collisions between ions and residual gas atoms result in the neutralisation of the dopant ion.

$$P^{+} + N_2 \rightarrow P^{\circ}\left(\text{neutral} + N_2^{+}\right)$$

- Neutrals cause non-uniform doping.
- Suppression of the neutral beam by bending the ion beam with electrostatic plates, leaves the neutral beam to travel straight ahead away from the wafers.

Beam Scanning

- For uniform doping, a beam is swept across the entire wafer in a creator scan pattern.
- Mechanical scanning helps holding the beam in one position and moving the wafer in front of it.

Crystal Damage

- Crystal structure is damaged by the colliding ions.
- There are three types of damage.
  - Lattice damage
  - Damage cluster
  - Vacancy interstitial
    - **Lattice damage:** ions collide with hot atoms and displace them from their lattice site.
    - **Damage cluster:** displaced atoms, in turn, displace other substrate atoms, creating a cluster of displaced atoms.
    - **Vacancy interstitial:** ion knocks a substrate atom from its lattice site and the displaced atoms come to rest in a non-lattice position.
- Light atoms, such as B, produce a small number of displaced atoms.
- Heavier atoms, such as P, As, generate a large number of displaced atoms.
- With prolonged bombardment, regions may change to an amorphous (non-crystal) structure.
- The damaged regions do not have the required electrical characteristics because the implanted atoms do not occupy lattice sites.

Annealing

- Restoration of the crystal damage and electrical activation of the dopants is achieved by the thermal heating step.
- Annealing temperature is less than diffusion temperature to prevent lateral diffusion.
- 600–1000°C for 15–30 min.

Deposition

CVD and Epitaxy

The basics of Chemical Vapour Deposition (CVD) are as follows.

1. In SC processing, deposition refers to a process, in which a material is deposited intact on the wafer surface as opposed to the formation of films by a surface chemical reaction, such as SiO_2.
2. Majority of films are deposited by CVD techniques.
3. Chemicals containing the atoms required in the film are mixed and reacted in a deposition chamber. The reaction forms the proper film elements in vapour state. The element or molecules deposit on the wafer surface and build up to form a film.

4.

Pyrolysis	$SiH_4 = Si + 2H_2$	examples of CVD reactions
Reduction	$SiCl_4 + 2H_2 = Si + 4HCL$	
Oxidation	$SiH_4 + O_2 = SiO_2 + 2H_2$	
Nitridation	$3SiH_2Cl_2 + 4NH_3 \rightarrow Si_3N_4 + pH + 6H_2$	

5. Deposited film growth proceeds in several distinct stages.
   a. First nucleation—first few atoms deposit on the surface.
   b. These atoms form islands that grow into larger islands.
   c. Islands spread into a continuous film. This is the transition stage.

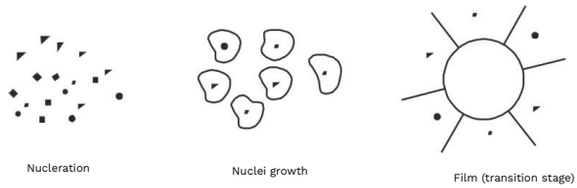


Fig. 4.29 CVD Film Growth Steps

Systems Overview

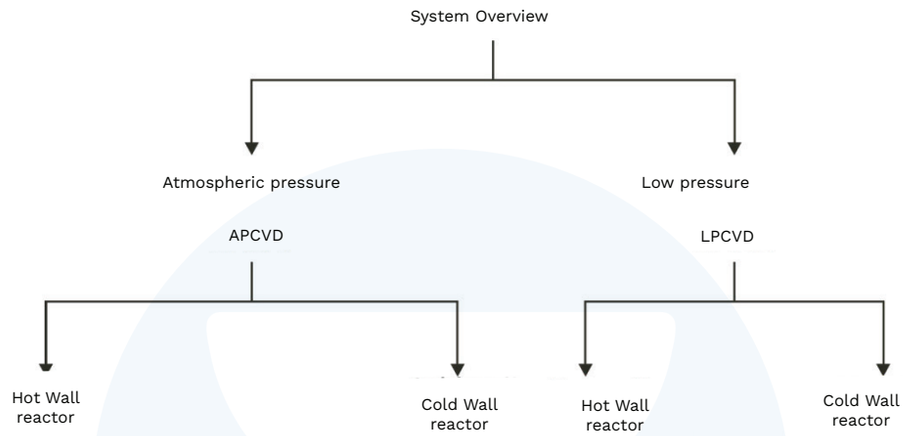


Fig. 4.30

Cold Wall

- It directly heats the wafer holder or wafer through radiant heating.
- The reaction occurs only at the heated wafer holder.

Hot Wall

- It heats up the wafer, holder, and chamber.
- There is reaction throughout the chamber.
- It necessitates frequently cleaning to avoid contamination.

PECVD

- There is plasma enhanced CVD.
- There is enhanced film deposition.
- There is uniform deposition at lower temperatures.
- There is low pressure, low temperature, good film uniformity, and throughput.

LPCVD

- A factor influencing film uniformity is the mean free path.
- Mean free path is the average distance a molecule travels before collision.
- Longer the mean free path, higher is the uniformity.
- Lowering the pressure increases the mean free path.
- It also allows lower deposition temperature.

Advantages

- It needs a lower chemical reaction temperature.
- It has a good step coverage and uniformity.

Epitaxial Silicon

1. Epitaxy means ‘arranged upon.’
2. The depositing atoms arrive at the wafer surface with sufficient energy to move around on the surface and orient themselves to the crystal arrangement of the wafer atoms.

Thus, an epitaxial film deposited on a < 111 > oriented wafer will take on a < 111 > orientation.

3. Si tetrachloride $SiCl_4 + 2H_2 \longleftrightarrow Si + 4HCL$

   Silane $SiH_4 + heat \longleftrightarrow Si + 2H_2$

   Dichlorosilane $SiH_2Cl_2 \longleftrightarrow Si + 2HCL$

   Epitaxial Si Chemical sources

4. Its advantages are as follows.
   a. It has a precise doping and doping range.
   b. Si wafers are manufactured in a concentration range of approximately 10^{13} to 10^{19} atoms/cm^3.
   c. Epitaxial films can be grown from 10^{12}–10^{20} atoms/cm^3.
5. Epitaxial film in bipolar technology is an n-type epitaxial film on a p-type wafer (NPN and PNP).
6. The faults include spikes, stacking faults, and pattern shift slip.

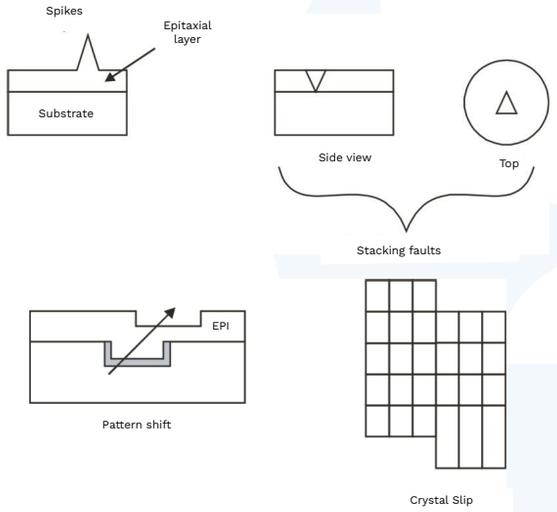


Fig. 4.31 Faults

CMOS Epitaxy:

Earlier, the dominant use of epitaxial films was the collector region of bipolar transistors. The technique provided a quality substrate for the device operation and a clever means of isolating adjacent devices.

The need for an epitaxial layer was driven by a CMOS circuit problem called latch-up.

The solution to prevent latch-up is a low-resistivity epitaxial layer that shunts (shorts) the emitter of the bipolar transistor so that it does not turn on.

Selective Epitaxial Silicon

Advancement in epitaxial deposition systems has introduced the selective growth of epitaxial films. The epitaxial films for bipolar and CMOS substrates are deposited on the entire wafer in selective growth. They are grown through holes in either silicon dioxide or silicon nitride films.

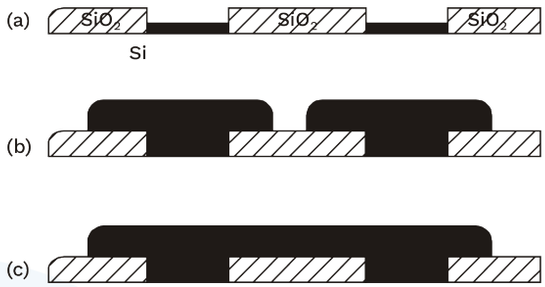


Fig. 4.32 Steps in Selective Epitaxial Growth

An advantage of such a structure is that the devices formed on the surfaces of the epitaxial regions are isolated from each other by the oxide or nitride regions.

Molecular Beam Epitaxy (MBE)

It is an evaporation process. The deposition chamber is maintained at a low pressure of 10^{-10} torr. The effusion cells contain a pure sample of target material desired on the wafer.

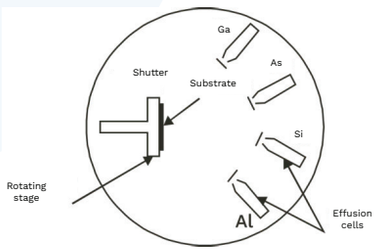


Fig. 4.33 Diagram of MBE Deposition System

Process: an electron beam is directed into the centre of the target material, which heats it to the liquid state. Atoms evaporate out of the material, exit the cell, and deposit on the wafers.

Advantages

- Low temperature (400–800°C) is required, which minimises auto doping and out diffusion.
- There is the ability to form multiple layers on the wafer surface during one process step.

Disadvantage

Low film growth rate 60–600 A°/min.

Vapour Phase Epitaxy (VPE)

A VPE system is a combination of a standard liquid source tube furnace and a two zone diffusion furnaces.

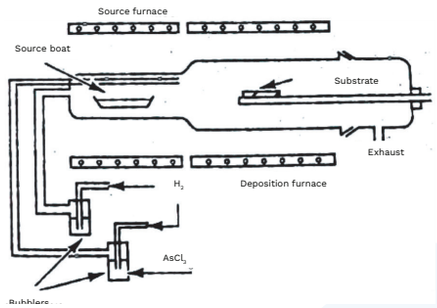


Fig. 4.34 Diagram of Gallium Arsenide VPE Deposition System

VPE refers to a compound material deposition system. An example of a particular arrangement in Fig. 4.34 is used to deposit epitaxial gallium arsenide. The creation of GaAs layer on wafer proceeds in two stages. Arsenic trichloride reacts with hydrogen in the first section to form arsenic.

$$4AsCl_3 + 6H_2 \rightarrow 12HCL + As_4$$

Arsenic deposits on the gallium form a crust. Hydrogen passing over the crust reacts in the first section to form three gases that pass into the wafer section.

$$\underset{(solid)}{GaAs} + \underset{(gas)}{HCL} \leftrightarrow \underset{(gas)}{GaCl} + \underset{(gas)}{\frac{1}{2}H_2} + \underset{(gas)}{\frac{1}{4}As_4}$$

This section is at lower temperature and the reaction proceeds in reverse, depositing GaAs on the wafers. The technique offers the advantage of clean films and has higher production rates than MBE technique but the film structure produced does not match the quality of MBE films.

Metallization

Metallization connects the various parts of a device and serves as electrical terminals for the chip.

Metal Film Uses

Conductors: the most common and familiar use of metal films in semiconductor technology is for surface wiring. The process of wiring the components parts together is generally referred to as metallization.

The metallization process starts in the masking area, where small holes called contact holes or contacts are etched through the surface layers down to the active regions of the devices. Following contact masking, a thin layer (10000 to 15000 A°) of the conducting metal is deposited by vacuum evaporation, sputtering, or CVD techniques over the entire wafer. This step leaves the surface covered with thin lines of metal called leads, metal lines, or interconnects. With increased chip density, the area available for surface wiring decreases, and thus, multilevel schemes start.

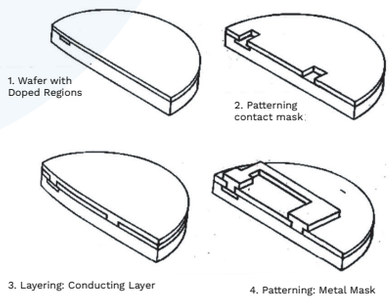


Fig. 4.35 Metallization Sequence

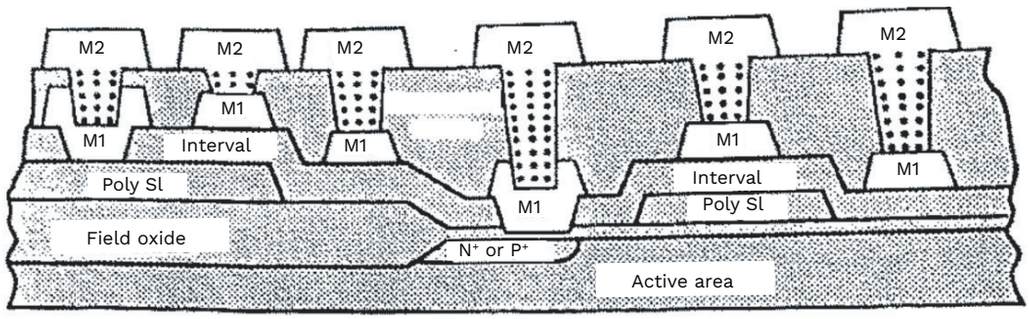


Fig. 4.36 Cross Section of Typical Planarised Two-level Metal VLSI Structure Showing Grange via Depths after Planarisation

Fuses: the development of thin film fuse technology allows the creation of the programmable read only memory (PROM) circuit. The fuse allows field programming of data in the memory section of the chip. In this role, the fuse is included specifically to be blown or disabled. Films of nichrome or titanium–tungsten are used for fuse materials.

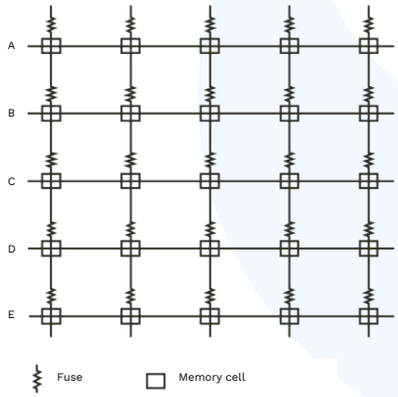


Fig. 4.37 Schematic of Fused Memory Array

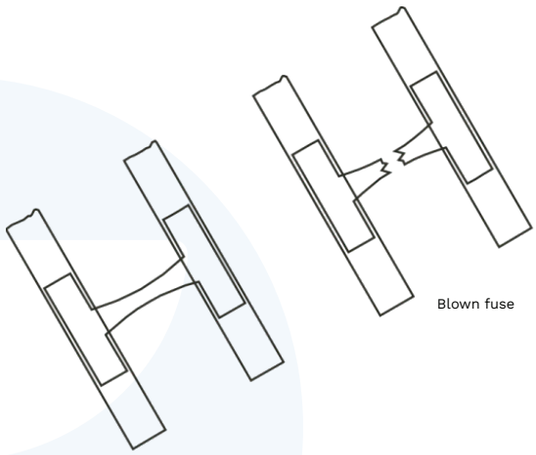
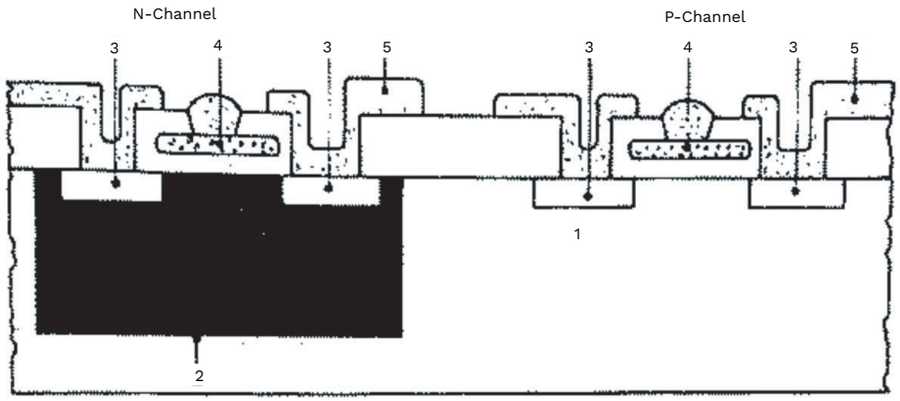


Fig. 4.38 Thin Film Fuses

MOS gate electrode: in MOS transistors, a voltage is applied to a metal lead over the gate region. The gate is formed from deposited polysilicon. The doped polysilicon functions as an MOS gate electrode and as a circuit conductor.



1. Wafer
2. P-Well
3. Source/drain
4. Gate
5. Gate oxide
6. Metallization

Fig. 4.39 CMOS Transistor Polysilicon Gate Electrode

Intermediate Barrier and Protective Films

Advanced circuit designs require the use of metal film sandwiches. The sandwiches of two or three different metals take the advantage of some properties of one of the metals. The other ones may function as an intermediate layer between the wafer and the second metal. One may serve to separate two incompatible metals or separate a metal and a semiconductor.

Others may serve to protect a lower level metal from physical abuse.

Backside plating: gold on the backside of wafer acts as a solder in certain packaging processes.

Metal Materials

Aluminium

- Aluminium is less conductive than copper and gold but copper has high contact resistance with silicon and its unacceptability inside silicon devices makes it inappropriate.
- Gold makes poor direct electrical contact with silicon. Gold causes leakage of current that destroys device functioning. Another problem with gold is its softness, which requires an over layer of molybdenum for protection.
- So, aluminium emerges as the preferred metal because it has the following properties.
  - Low enough resistivity (27 micro ohm cm)
  - Good current carrying density
  - Superior adhesion to SiO_2
  - Available in high purity
  - Naturally low contact resistance with silicon
  - Relatively easy to pattern with conventional PR processes

Aluminium Silicon Alloy

- Aluminium and silicon start dissolving into each other while heating and lead to eutectic formation point.
- An eutectic formation occurs when two materials are heated lower than their individual melting temperatures. So, two solutions are employed to this problem.
  - Barrier metal layer between aluminium and silicon.
  - Alloy of aluminium with 1 to 2 per cent silicon is used rather than pure aluminium. During the contact heating step, aluminium alloys with more silicon in the alloy itself is used and the silicon from wafer is used less.

Aluminium Copper Alloy

- Aluminium suffers from a problem called electro migration. The problem occurs in current carrying skinny aluminium leads over long distances. The current sets up an EF and the heat generated by current sets up a thermal gradient along the lead. The aluminium in the lead becomes mobile and diffuses within itself along the direction of the two gradients. The first effect is thinning of the lead. In the extreme conditions, the lead can become completely separated.
- So, an alloy of aluminium and 4 per cent copper, or an alloy of aluminium and 0.1 to 0.5 per cent titanium is preferred to be used.
- A drawback of using aluminium alloys is that it increases complexity for deposition and increases film resistivity.

Doped Polysilicon

- It is used in silicon gate MOS technology. The polysilicon is doped to increase its conductivity. The commonly preferred dopant is phosphorous due to its high solid solubility in silicon.
- Doping is either done by diffusion, ion implantation, or in situ doping during the Low Pressure Chemical Vapour Deposition (LPCVD) process. Each method produces different doping results.
- The lower the temperature, the greater the amount of dopants trapped in the polygrain structure, which is unavailable for conduction. This is the situation with ion implantation.
- Diffusion doping results in the lowest film sheet resistivity.
- In situ, CVD doping has the lowest dopant carrier mobility due to grain boundary trapping.

- Polysilicon has a lower contact resistance than aluminium and doped polysilicon makes good ohmic contact with silicon wafer.

Titanium–Tungsten Barriers

To prevent eutectic alloying of silicon and aluminium, a layer of titanium–tungsten (TiW) is sputter deposited onto the wafer into the open contacts before aluminium deposition takes place.

Refractory Metals and Refractory Metal Silicides

- Refractory metals and their silicides offer lower contact resistance.
- Refractory metals of interest are titanium (Ti), tungsten (W), tantalum (Ta), and molybdenum (Mo).
- Their silicides form (they are alloyed on a silicon wafer) WSi_2, $TaSi_2$, $MoSi_2$, and $TiSi_2$
- They were first proposed for metallization in 1950s but stayed in the background due to the lack of reliable deposition methods. The situation changed with the development of LPCVD and sputtering processes.

Deposition Methods

- The mainstay of metal deposition up to the mid-1970s was vacuum evaporation. Aluminium, gold, and fuse metals were deposited by this technique.
- The need to deposit multi-metal systems and alloys, along with the need for better step coverage, led to the introduction of sputtering technique. Refractory metal use has added the third technique Chemical Vapor Deposition (CVD).

Metallization Problems

There are various metallization problems associated with deposition, processing, metallurgical and chemical interactions, electro migration, device performance, etc.

Deposition

Problems encountered in deposition include impurities in the film, adhesion, stress, cracks, grain sizes, stoichiometry in alloys, step coverage, and thickness non-uniformity.

Impurities can be minimised by using pure evaporation and sputtering sources, high vacuum and pure gases, and clean surfaces. Adhesion can be improved by surface cleanliness and localised deposit surface interaction.

To have the lowest stress (thermal and intrinsic stress) conditions in the deposits, sputter pressure stress, and sputter rate stress relationships need to be considered.

Cracks in films are because of poor adhesion, particulates, or unevenness. Larger grains provide more stability.

Step coverage is generally found in physical vapour deposited films because of directionality of the deposition, low mobility of deposited atoms, and enhanced topography. The solutions to step coverage are as follows.

- Raising the substrate temperature during deposition to about 300°C as it increases the surface mobility of the deposited material, thus, reducing the possibility of cracks
- Optimising the orientation of the substrate relative to the source

Processing

The gate metal is deposited earlier than the contact and top level metal. Thus, the gate metal is subjected to considerable processing, whereas contact metal experiences little processing.

Several high temperature and ion implantation steps lead to stability problems. Polysilicon alone has been used to solve these problems. Nowadays, polysilicon silicide sandwich has replaced polysilicon because the stability of this sandwich during processing and device operations is better than polysilicon. Silicides of titanium and tantalum, tungsten and molybdenum, and titanium and cobalt are generally used.

Metallurgical and Chemical Interactions

Although pure silicide films on silicon polysilicon are stable. Sometimes, they are slightly affected by some chemical solutions used in IC fabrication. For example, $TiSi_2$ dissolves quickly in HF containing solutions. So, exposure of $TiSi_2$ to such solutions needs to be avoided.

Aluminium on silicon or silicide leads to metallurgical interactions that cause serious instabilities. Annealing of aluminium of silicon causes dissolution of silicon by diffusion into the metal and leads to pit formation. Contact failures, in Al–Si systems, also occur due to precipitation of dissolved silicon (from aluminium) on cooling, which causes unwanted increases in contact resistance.

To solve the problems of Al–Si interaction, a diffusion barrier between silicon and aluminium is used. Self-aligned silicides of Pt, Pd, Co, Ni, and Ti are commonly used as diffusion barriers.

Practice Exercise

Level I

1. Which process is preferred to form gate dielectric (SiO_2) of MOSFETs?
(A) Sputtering
(B) CVD
(C) Wet oxidation
(D) Dry oxidation

2. Consider the following processes involved in IC fabrication.
I. Oxidation
II. Diffusion
III. Photolithography
IV. Metallization
The correct sequence of steps is:
(A) I, III, II, and IV
(B) IV, III, I, and II
(C) I, II, IV, and III
(D) I, II, III, and IV

3. If P is development, Q is etching, R is alignment and exposure, and S is photoresist stripping, then their order in IC fabrication process should be:
(A) P, Q, R, and S
(B) R, P, Q, and S
(C) P, R, Q, and S
(D) S, R, P, and Q

4. The _____ is used to reduce the resistivity of polysilicon.
(A) Photoresist
(B) Etching
(C) Doping impurity
(D) None of the above

5. The isolated active areas are created by a technique known as:
(A) etched field oxide isolation
(B) local oxidation of silicon
(C) DIBL
(D) development

6. Photoresist is formed using:
(A) high sensitive polymer
(B) light sensitive polymer
(C) polysilicon
(D) None of the above

7. In CMOS fabrication, the photoresist is exposed to:
(A) infra-red light
(B) ultra violet light
(C) visible light
(D) fluorescent light

8. Which type of CMOS circuits is generally preferred?
(A) P-well
(B) N-well
(C) Twin tub
(D) None of the above

9. Etching refers to the removal of material from:
(A) soft surface
(B) sticky surface
(C) wafer surface
(D) hard surface

10. In wet etching, material is removed by:
(A) absorption
(B) sublimation

(C) chemical reaction
(D) force exerted due to the flow of solvent

11. _____ is used to protect the remaining area of the wafer while etching.
(A) Tin foil
(B) Wood
(C) Photoresist
(D) Metal

12. In plasma etching, chemical etchant is used in:
(A) liquid phase
(B) gas phase
(C) powder form
(D) gel form

13. Which of the following is a combination of other two processes?
(A) Plasma etching
(B) Wet etching
(C) Ion beam etching
(D) Reactive ion etching

14. System on chip means:
(A) analog IC
(B) digital IC
(C) Both (A) and (B)
(D) complex IC

15. Which of these is an advantage of IC?
(A) Small size
(B) Low cost
(C) High packing density
(D) All of the above

16. Monolithic ICs are:
(A) forms of discrete circuits
(B) on a single chip
(C) a combination of thin and thick film circuit
(D) also called hybrid ICs

17. Which of these is a problem associated with diffusion?
(A) Lateral diffusion
(B) Dislocation generation
(C) Low doping control
(D) All of the above

18. Ion implantation is a:
(A) physical process
(B) chemical process
(C) electrochemical process
(D) thermal process

19. Ion implantation process takes place at:
(A) Room temperature
(B) Close to room temperature
(C) High temperature
(D) Below room temperature

20. PECVD refers to:
(A) Pressure Enhanced CVD
(B) Plasma Enhanced CVD
(C) Pressure End CVD
(D) None of the above

21. Epitaxy means:
(A) arranged upon
(B) highly concentrated
(C) thin
(D) thick

22. Which of the following is an epitaxial silicon chemical source?
(A) Silicon tetrachloride
(B) Silane
(C) Dichlorosilane
(D) All of the above

23. Which of the following is a fault in an epitaxial grown film?
(A) Spikes
(B) Stacking faults
(C) Pattern shift
(D) All of the above

24. In plasma enhanced CVD, deposition occurs at a _____ temperature.
(A) higher
(B) constant
(C) increasing
(D) lower

25. Which one of the following is not a strengthening mechanism?
(A) Alloying
(B) Baking
(C) Annealing
(D) Metallising

26. Etching is used for:
(A) selective removal of unwanted surface
(B) cleaning
(C) interconnection
(D) All of the above

27. Metallization is used for:
(A) interconnection
(B) cleaning
(C) deposition
(D) None of the above

28. Alloying solves the problem of:
(A) electro migration
(B) thermal gradient
(C) high voltage
(D) melting

29. During the ion implantation process (before the ion strike the wafer), the accelerated ions are passed through:
(A) strong electric field
(B) strong magnetic field
(C) Both (A) and (B)
(D) None of the above

30. Shallow penetration of dopants is possible in:
(A) ion implantation
(B) vertical diffusion
(C) horizontal diffusion
(D) dopants diffusion

31. The interconnections are made during:
(A) diffusion process
(B) lithography process
(C) epitaxial growth
(D) metallization process

32. Optical masking is used for:
(A) safety
(B) cleaning
(C) pattern transfer
(D) None of the above

Level II

1. The oxidation rate in an oxidation process:
(A) depends on the oxide thickness grown and temperature
(B) depends on temperature only
(C) slows down as the oxide grows
(D) is independent of temperature

2. Which of the statements are correct about dry oxidation?
I. It is used to form gate oxides of MOSFETs.
II. It has better quality oxide than obtained by wet oxidation.
III. Poor quality oxide is grown by this method.
IV. It is slower than wet oxidation.
(A) I, II, and IV only
(B) I, III, and IV only
(C) III and IV only
(D) IV only

3. For growing thick oxides:
(A) dry oxidation is preferred
(B) wet oxidation is preferred
(C) dry followed by dry oxidation
(D) wet followed by dry oxidation

4. In IC technology, dry oxidation as compared to wet oxidation process has:
(A) better quality oxide with a higher growth rate
(B) inferior quality oxide with a higher growth rate
(C) better quality oxide with a slower growth rate
(D) inferior quality oxide with a lower growth rate

5. Lithography is a process to:
(A) transfer a pattern to a layer on the chip
(B) develop an oxide layer on the chip
(C) develop a metal layer on the chip
(D) define the active area on the chip

6. Positive photoresists are used more than negative photoresists because:
(A) their photolithographic resolution is higher than negative photoresists
(B) positive photoresists are more sensitive to light
(C) positive photoresists are less sensitive to light
(D) negative photoresists are less sensitive to light

7. Which of the following holds true for wet etching?
(A) It is used for large pattern sizes.
(B) It is an anisotropic process.
(C) It forms straight sidewalls.
(D) Undercutting and resisting peel off can happen if time is not controlled.

8. In dry etching, gases are:
(A) absorbed and react with the material
(B) adsorbed and react with the material
(C) used to remove the burr
(D) given high velocity along with abrasive particles, which will remove the material

9. For the fabrication of ICs, silicon is used because:
(A) silicon is mechanically strong
(B) silicon is a better conductive semiconductor material
(C) silicon possesses characteristics, which are best suited for IC manufacturing processes
(D) silicon is cheaper

10. In ion implantation:
(A) there is lateral diffusion
(B) the process takes place at a high temperature
(C) wide range of doping concentration is possible
(D) dopant atoms are placed above the wafer surface

11. In ion implantation method, penetrating the ions into the silicon wafer depends on:
(A) accelerating voltage
(B) accelerating speed
(C) accelerating current
(D) all of the above

12. Which are the statements are correct regarding epitaxial growth?
I. Thin layers are grown on a substrate wafer; this technique is known as epitaxial growth.
II. PVD is also called VPE.
III. OMCVD is a technique to grow epitaxial layers using metal organic compounds.
IV. Slow deposition rate is its disadvantage.
(A) I, II, and III only
(B) II and IV only
(C) I, II, and IV only
(D) I and III only

13. Which of these statements are true about refractory metal and their silicides.
I. They offer low contact resistance.
II. They are deposited by LPCVD technique.
III. They are deposited by sputtering process.
IV. They are deposited at low temperatures.
(A) I, II, and IV only
(B) I, II, and III only
(C) III and IV only
(D) I and IV only

14. The silicon dioxide layer in an IC acts as:
(A) an insulating layer
(B) a conducting layer
(C) barrier in the selective diffusion of impurities in epitaxial layer and protects
(D) Both (A) and (C)

15. How the aluminium film coating is carried out in metallization process?
(A) Heating and pouring aluminium in required place
(B) Aluminium is vacuum evaporated, and then, condensed
(C) Placing the aluminium in required place, and then, heating it using tungsten
(D) None of the above

Answer Key

Level I

1. (D)	**2.** (A)	**3.** (B)	**4.** (C)	**5.** (B)	**6.** (B)	**7.** (B)	**8.** (B)	**9.** (C)	**10.** (C)
11. (C)	**12.** (B)	**13.** (D)	**14.** (C)	**15.** (D)	**16.** (B)	**17.** (D)	**18.** (A)	**19.** (B)	**20.** (B)
21. (A)	**22.** (D)	**23.** (D)	**24.** (D)	**25.** (D)	**26.** (A)	**27.** (A)	**28.** (A)	**29.** (B)	**30.** (A)
31. (D)	**32.** (C)								

Level II

1. (C)	**2.** (A)	**3.** (B)	**4.** (C)	**5.** (A)	**6.** (A)	**7.** (D)	**8.** (B)	**9.** (C)	**10.** (C)
11. (A)	**12.** (A)	**13.** (B)	**14.** (D)	**15.** (B)					